METRIC

DOD-W-70336
31 December 1987

MILITARY SPECIFICATION

WASHERS, LOCK, (METRIC), GENERAL SPECIFICATION FOR

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers metrically dimensioned carbon steel, corrosion resistant and phosphor bronze lock washers, helical spring, internal tooth and external tooth.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

QQ-P-416 - Plating, Cadmium (Electrodeposited)
QQ-S-763 - Steel Bars, Wire, Shapes and Forgings, Corrosion-Resisting
PPP-H-1581 - Hardware (Fasteners and Related Items), Packaging of

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research, Development and Engineering Center, ATTN: SMCAR-ESC-S, Picatinny Arsenal, NJ 07806-5000 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 5310

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MILITARY

MIL-T-10727 - Tin Plating; Electrodeposited or Hot-Dipped, for Ferrous and Non Ferrous Metals
MIL-C-13924 - Coating, Oxide, Black, for Ferrous Metals
MIL-I-17214 - Indicator, Permeability; Low-Mu (GO - NO GO)
MIL-P-47158 - Penetrant Inspection, Soundness Requirements for Materials, Parts and Weldments
DOD-W-70336/1 - Washer, Lock Spring, Helical, Regular Series (Metric)
DOD-W-70336/2 - Washer, Lock Spring, Helical, Light Series (Metric)
DOD-W-70336/3 - Washer, Lock, Flat-Internal Tooth (Metric)
DOD-W-70336/4 - Washer, Lock, Flat-External Tooth (Metric)

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-1312 - Fastener Test Methods
MIL-STD-1949 - Inspection, Magnetic Particle

(Copies of specifications, standards and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issue of the non-Government documents which is current on the date of the solicitation.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI/ASME B46.1 - Surface Texture (Surface Roughness, Waviness and Lay)

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, NY 10018-3363.)

ASTM

ASTM A510M - General Requirements for Wire Rods and Coarse Round Wire, Carbon Steel (Metric)

ASTM B139M - Phosphor Bronze Rod, Bar and Shapes (Metric)

ASTM A380 - Cleaning and Descaling Stainless Steel Parts, Equipment and Systems

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(Applications for copies should be addressed to ASTM, 1916 Race Street, Philadelphia, PA 19103-1187.)

(Non-Government standards and other publications are normally available from the organizaions which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Specification sheets. The individual item requirements shall be as specified herein and in accordance with the applicable specification sheet. In the event of any conflict between requirements of this specification and the specification sheet, the latter shall govern (see 6.2).

3.2 Material. Recycled and reclaimed materials shall be used to the maximum extent practicable. Washers shall be made of spring steel of suitable quality to meet the requirements specified herein.

3.2.1 Spring steel. Spring steel shall conform to any of the steel Grade 1050 thru 1095 (UNS G10500 thru UNS G10950), as specified in ASTM A510M.

3.2.2 Corrosion resistant steel. Austenitic corrosion resistant steel (CRES) shall conform to any of the types within the 300 series (UNS S3XXXX), as specified in QQ-S-763.

3.2.3 Phosphor bronze. Phosphor bronze shall conform to ASTM B139M (UNS S51000), temper H04, hard.

3.3 Hardness and finish.

3.3.1 Surface hardness.

3.3.1.1 Spring steel. Carbon steel helical spring lockwashers shall have a surface hardness of 45-52 HRC. Carbon steel lock-tooth washers shall have a surface hardness of 40-50 HRC.

3.3.1.2 Corrosion resistant steel. Austenitic corrosion resistant steel helical washers shall exhibit a surface hardness of 35-45 HRC.

3.3.1.3 Phosphor bronze. Phosphor bronze lock-tooth washers shall have a surface hardness of 85-95 HRB.

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3.3.2 Finish. Washers shall be plain (unplated or uncoated) with a light film of oil or rust inhibitor, cadmium plated (electrodeposted), tinned, black oxide coated or passivated as specified on the applicable specification sheet as indicated below.

3.3.2.1 Electrodeposited cadmium. Electrodeposited cadmium plated washers shall be in accordance with QQ-P-416, Type II, Class 2 (7.6 µm thick).

3.3.2.2 Black oxide. Corrosion resistant steel (CRES) washers shall be black oxide coated in accordance with MIL-C-13924, Class 4.

3.3.2.3 Cleaning, descaling, and passivating. Corrosion resistant steel (CRES) washers shall be cleaned, descaled, and passivated in accordance with ASTM A380.

3.3.2.4 Tinplate. Phosphor bronze washers shall be tinplated in accordance with MIL-T-10727, Type I.

3.4 Dimensions. Dimensions and tolerances shall be in accordance with the applicable specification sheet and shall apply after protective finish.

3.5 Edges and burrs. All sharp edges and burrs shall be removed.

3.6 Surface roughness. Washers shall have a maximum surface roughness of 3.2 µm in accordance with ANSI/ASME B46.1.

3.7 Decarburization. Carbon steel lockwashers shall be free from decarburization.

3.8 Cracks. Washers shall be free from cracks, flaws and pits in any location.

3.9 Magnetic permeability. The magnetic permeability of austenitic corrosion resistant steel washers shall not be greater than 2.0 (AIR = 1.257 µH/m) (micro henry per meter) at a field strength of 16 KA/m (kilo ampere per meter).

3.10 Performance.

3.10.1 Free height test. Helical spring washers shall withstand the free height-loading test specified in 4.4.6 without indications of permanent set or cracking. The measured free height after test shall be two-thirds of the original free height specified on the appropriate specification sheet.

3.10.2 Compression and flattening resistance test. Lock-tooth washers shall withstand the compression and flattening resistance test of 4.4.7 without evidence of fracture of the teeth or rim, nor shall the teeth be flattened with respect to the rim. The measured free height after test shall be two-thirds of the original free height specified on the appropriate specification sheet.

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3.11 Workmanship. Washers shall be free of surface contamination, tool marks and other imperfections which may adversely affect useability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specifiction where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Quality conformance inspection.

4.2.1 Inspection of product for delivery. Inspection of product for delivery shall consist of Group A and B inspections.

4.2.1.1 Inspection lot. An inspection lot shall consist of all washers covered by a single specification sheet, produced under essentially the same conditions, and offered for inspection at any one time.

4.2.1.2 Group A inspection. Group A inspection shall consist of the inspection specified in Table I. The Acceptable Quality Level (AQL) shall apply to each individual defect, not to a group of defects.

4.2.1.2.1 Sampling plan. Statistical sampling and inspection for Subgroup 1 shall be in accordance with MIL-STD-105, Inspection Level S-3. Major and minor defects shall be as defined in Table II. Sampling and inspection for Subgroup 2 shall be in accordance with MIL-STD-105, Inspection Level S-1.

4.2.1.2.1.1 Rejected lots (Subgroup 1). Rejected inspection lots may be resubmitted for Government acceptance only if the manufacturer performs 100 percent inspection on the washers of the lot for those characteristics which were defective and resulted in rejection of the lot and removes all defective units and resubmits the lot for quality conformance inspection. Resubmitted lots shall be kept separate from new lots and shall be clearly identified as resubmitted lots. Resubmitted lots shall be inspected using the tightened inspection procedure of MIL-STD-105 and shall not thereafter be tendered for acceptance unless the former rejection or requirement of correction is disclosed.

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TABLE I. GROUP A INSPECTION.

INSPECTION	REQUIREMENT PARAGRAPH	TEST METHOD PARAGRAPH	SAMPLING PROCEDURE	
Subgroup 1			AQL (Percent defective)	
			Major	Minor
Dimensions	3.4	4.4	2.5	4.0
Finish	3.3.2	4.4.3.2	--	4.0
Surface roughness	3.6	4.4.1	--	4.0
Subgroup 2			AQL (Percent defective)	
Magnetic permeability	3.9	4.4.3.3	2.5	
Cracks	3.8	4.4.1	1.5	

4.2.1.2.2 Disposition of sample units. Sample units which have passed all the Group A inspections may be delivered on the contract or purchase order.

TABLE II. CLASSIFICATION OF DEFECTS.

CATEGORY	DEFECT	INSPECTION METHOD
Critical	None defined	
Major	AQL = 2.5 percent defective	
101	Inside diameter, not as specified (see 3.4)	CIE 1/
102	Outside diameter, not as specified (see 3.4)	CIE 1/
103	Evidence of cracks or pits (see 3.8)	CIE 1/
104	Thickness, not as specified (see 3.4)	CIE 1/
105	Free height, not as specified (see 3.4)	CIE 1/
Minor	AQL = 4.0 percent defective	
201	Protective coating, missing or incomplete (see 3.3.2)	Visual
202	Other dimensions, not as specified (see 3.4)	CIE 1/
203	Surface roughness, not as specified (see 3.5)	CIE 1/
204	Workmanship (see 3.10)	Visual

1/ Commercial inspection equipment

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4.2.1.2.1.2 Rejected lots (Subgroup 2). If an inspection lot is rejected, the manufacturer may rework it to correct the defects, or screen out the defective units, and resubmit for reinspection. Resubmitted lots shall be inspected using tightened inspection procedure of MIL-STD-105 and shall not thereafter be tendered for acceptance unless the former rejection or requirement of correction is disclosed. Such lots shall be separate from new lots and shall be clearly identified as reinspected lots.

4.2.1.3 Group B inspection. Group B inspection shall consist of the tests specified in Table III in the order shown. They shall be performed on sample units that have been subjected to and have passed the Group A inspection.

TABLE III. GROUP B INSPECTION.

INSPECTION	REQUIREMENT PARAGRAPH	TEST METHOD PARAGRAPH	AQL (PERCENT DEFECTIVE)
Surface hardness	3.3.1	4.4.3.1	1.5
Decarburization	3.7	4.4.5	1.5
Free height	3.10.1	4.4.6	1.5
Compression and flattening	3.10.2	4.4.7	1.5

4.2.1.3.1 Sampling plan. The sampling plan shall be in accordance with MIL-STD-105. Unless otherwise specified herein, S-1 inspection shall be used.

4.2.1.3.2 Defectives. If the number of defects exceed the number allowed in Table II, the sample shall be considered to have failed.

4.2.1.3.3 Disposition of samples. Sample units which have been subjected to Group B inspection shall not be delivered on the contract or purchase order.

4.2.1.3.4 Noncompliance. If a sample fails to pass Group B inspection, the manufacturer shall notify the qualifying activity and the cognizant inspection activity of such failure and take corrective action on the materials or processes, or both, as warranted, and on all units of product which can be corrected and which are manufactured under essentially the same materials and processes, and which are considered subject to the same failure. Acceptance and shipment of the product shall be discontinued until corrective action acceptable to the qualifying activity has been taken. After the corrective action has been taken Group B inspection shall be repeated on additional sample units (all tests and examinations, or the test which the original sample failed, at the option of the qualifying activity). Group A inspection may be reinstated; however, final acceptance and shipment shall be withheld until the Group B inspection has shown that the corrective action was successful. In the event of failure after reinspection, information concerning the failure shall be furnished to the cognizant inspection activity and the qualifying activity.

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4.3 Inspection of packaging. Sampling and inspection of the preservation-packaging, packing and container marking shall be in accordance with the requirements of PPP-H-1581.

4.4 Methods of inspection.

4.4.1 Visual and dimensional. The washers shall be examined to verify that physical dimensions, surface roughness and workmanship are in accordance with the applicable requirements of 3.4, 3.5, 3.9 and 4.2.1.2.

4.4.2 Material inspection. Material inspection shall consist of certification supporting verifying data that the materials used in fabricating the washers are in accordance with the applicable requirements of 3.2.

4.4.3 Hardness and finish inspection.

4.4.3.1 Hardness inspection. Samples taken as specified in 4.2.1.3 shall be tested for surface hardness in accordance with MIL-STD-1312, Test Method 6 and 3.3.1.

4.4.3.2 Finish inspection. Samples taken as specified in 4.2.1.2 shall be inspected for adequacy of plating in accordance with applicable specification of 3.3.2.

4.4.3.3 Magnetic particle (cracks) inspection. Carbon steel washer samples taken as specified in 4.2.1.2 shall be subjected to magnetic particle inspection in accordance with MIL-STD-1949. Corrosion resistant steel (CRES) 300 Series austenitic and phosphor bronze samples, selected per 4.2.1.2, shall be subjected to penetrant inspection in accordance with MIL-P-47158. There shall be no evidence of cracks or pits as specified in 3.8.

4.4.4 Magnetic permeability inspection. Corrosion resistant steel (CRES) washers selected in accordance with 4.2.1.2 shall be tested for compliance with the magnetic permeability requirement of 3.9. The test shall be conducted in accordance with MIL-I-17214.

4.4.5 Decarburization. Samples taken as specified in 4.2.1.3 shall be microscopically examined at a magnification of 100 diameters. There shall be no evidence of decarburization, the etchant shall be 5 percent nital.

4.4.6 Free height test. Helical spring washers selected in accordance with 4.2.1.3 shall be subjected to a compression, twice pressed flat, released, and the free height measured. The free height after the second compression shall conform to the requirements of 3.10.1.

4.4.7 Compression and flattening test. Lock tooth washers selected in accordance with 4.2.1.3 shall be compressed flat and held for twelve hours. Upon release, the free height shall conform to the requirements of 3.10.2. There shall be no evidence of fracture of teeth or rim, or flattening of teeth with respect to rim when examined in accordance with 4.4.1.

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5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with PPP-H-1581 (see 6.2).

6. NOTES

6.1 Intended use.

Helical spring lock washers. The helical spring lock washers covered by this standard are intended for general industrial application. They have the function of: (1) providing greater bolt tension per unit of applied torque for tighter assemblies; (2) providing hardened bearing surfaces to create more uniform torque control; (3) providing uniform load distribution through controlled radii--section-- cut-off; and (4) providing protection against looseness resulting from vibration and corrosion.

Tooth lock washers. The tooth lock washers in this standard are intended for general industrial application. They serve to promote the retention of fasteners, such as bolts, screws and nuts, to the component parts of an assembly by increasing friction between the fasteners and the assembly and by providing spring takeup.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number and date of this specification and the applicable specification sheet.

b. Applicable specification part number (see 3.1).

c. Level (degree) of protection in accordance with PPP-H-1581, ordering data (see 5.1).

6.3 Subject term (key word) listing.

Washer, Lock Spring, Helical, Metric
Washer, Lock Spring, Helical, Regular Series
Washer, Lock Spring, Helical, Light Series
Washer, Lock, Metric
Washer, Lock, Flat-Internal Tooth, Metric
Washer, Lock, Flat-External Tooth, Metric

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Navy - OS
Air Force - 99

Review Activities:
Army - AV
Navy - SH
Air Force - 82
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Preparing Activity:
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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
DOD-W-70336	WASHERS, LOCK, (METRIC), GENERAL SPECIFICATION FOR

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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